

# BOARD QUESTION PAPER : MARCH 2017

## Notes:

- All questions are compulsory.
- Figures to the right indicate full marks.
- Answer to every question must be written on a new page.
- L.P.P. problem should be solved on graph paper.
- Log table will be provided on request.
- Write answers of Section – I and Section – II in one answer book.

## Section – I

### Q.1. Attempt any SIX of the following:

[12]

- Find  $x, y, z, w$  if 
$$\begin{bmatrix} x+y & x-y \\ y+z+w & 2w-z \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 9 & 5 \end{bmatrix}$$
 (2)
- Express the truth of the following statements with the help of Venn diagrams:
  - No circles are polygon
  - If a quadrilateral is rhombus, then it is a parallelogram. (2)
- Find the points of discontinuity, if any for the function:
$$f(x) = \frac{x^2 - 9}{\sin x - 9}$$
 (2)
- Write negation of the following statements:
  - The number 6 is an even number or the number 25 is a perfect square.
  - If  $x \in A \cap B$ , then  $x \in A$  and  $x \in B$  (2)
- Evaluate :  $\int \cos^2 x \cdot dx$  (2)
- Find  $\frac{d^2y}{dx^2}$ , if  $y = \log x$ . (2)
- Evaluate :  $\int \frac{e^x + 1}{e^x + x} \cdot dx$  (2)
- Find  $\frac{dy}{dx}$ , if  $x^3 + y^2 + xy = 10$  (2)

### Q.2. (A) Attempt any TWO of the following:

[6][14]

- Find the inverse of the matrix  $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 5 \\ 2 & 4 & 7 \end{bmatrix}$  by adjoint method. (3)
- If  $f(x) = \frac{e^{2x} - 1}{ax}$ , for  $x < 0, a \neq 0$   
 $= 1$ , for  $x = 0$   
 $= \frac{\log(1+7x)}{bx}$ , for  $x > 0, b \neq 0$   
is continuous at  $x = 0$ , then find  $a$  and  $b$ . (3)

- iii. Demand function  $x$ , for a certain commodity is given as  $x = 200 - 4p$  where  $p$  is the unit price.  
Find : a. elasticity of demand as function of  $p$ .  
b. elasticity of demand when  $p = 10$ , interpret your result. (3)

**(B) Attempt any TWO of the following:** [8]

- i. Using the truth table verify that  
 $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$ . (4)
- ii. If the demand function is  $D = 150 - p^2 - 3p$ , find marginal revenue, average revenue and elasticity of demand for price  $p = 3$ . (4)

- iii. Evaluate:  $\int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x}{1 + \sin^4 x} \cdot dx$ . (4)

**Q.3. (A) Attempt any TWO of the following:** [6][14]

- i. Solve the following equations by reduction method:  
 $x + 3y + 3z = 16$   
 $x + 4y + 4z = 21$   
 $x + 3y + 4z = 19$  (3)
- ii. If the function  
 $f(x) = \frac{15^x - 3^x - 5^x + 1}{x \tan x}$ ,  $x \neq 0$  is continuous at  $x = 0$ , then find  $f(0)$ . (3)
- iii. Examine the function  $f(x) = x + \frac{25}{x}$  for maxima and minima (3)

**(B) Attempt any TWO of the following:** [8]

- i. Find the volume of a solid obtained by the complete revolution of the ellipse  $\frac{x^2}{36} + \frac{y^2}{25} = 1$  about  $X - \text{axis}$ . (4)
- ii. If  $x^3 y^5 = (x + y)^8$ , then show that  $\frac{dy}{dx} = \frac{y}{x}$  (4)
- iii. Evaluate :  $\int \frac{(1 + \log x)}{x(2 + \log x)(3 + \log x)} \cdot dx$  (4)

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